

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

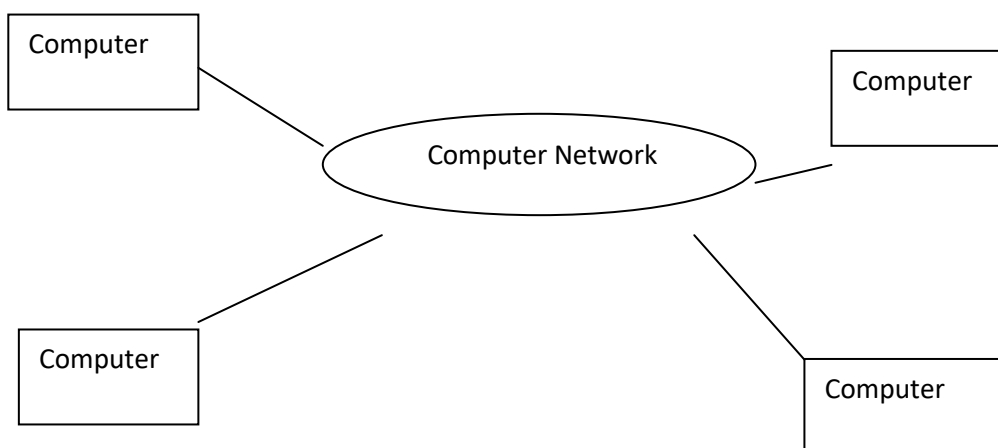
Chapter - 1

1. Computer Network Introduction:

Computer network is a system in which multiple computers are connected to each other to share information and resources.

A physical connection between networked computing devices is established using either cable media or wireless media.

The best known computer network is the INTERNET.



Advantages of Computer network:

1. **File Sharing:** The major advantage of a computer network is that allows file sharing and remote file access. A person sitting at one workstation that is connected to a network can easily see files present on another workstation, provided he is authorised to do so.
2. **Resource Sharing:** Resource Sharing is one of the important applications of Computer Networking. You can share single software among multiple users. We can also share Hardware Devices via this technique.
3. **Connectivity:** Computer Networks improve connectivity irrespective of a person's location. In these testing times, we can see live examples as we use video call apps or Google documents to connect with our friends and colleagues.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

4. **Reduced Cost:** Cost is one of the crucial factors that one needs to consider while evaluating the pros and cons of a particular technology. In networking, a central server is used that enables companies to store files in one place and thus reducing file storage expenses.
5. **Flexibility:** Here, flexibility means that different people will be able to explore different things as per their requirements. For this purpose, computer networks provide you a wide array of choices to share a particular piece of information. For example, e-mail or messaging apps like Whatsapp. So, there is flexibility for different users.

Disadvantages of computer networks

1. Lack of robustness
2. Spread of computer virus
3. Implementation issues
4. Lack of productivity
5. Health issues

1. Lack of robustness:

Computer networks rely on the main server called the central server. If the central server malfunctions or there is an issue in the central server, then the entire network would stop functioning. So, this is a major disadvantage due to dependency on a single server.

2. Spread of computer virus

As computers in a network are interconnected, there is a high probability that if one of the computers is affected by the virus, others too can get affected. This spread can actually damage the entire system. Also, if the central server gets corrupted, then it's quite dangerous as the network depends on the central server.

3. Independence issues

The entire networking system is based on a centralized server, so it lacks independence. So, individual users lack the freedom to use the computer as per their wish.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

4. Lack of productivity

Since a network has a lot of advantages and applications, it is certain that it results in simultaneous use of many services that cause distraction. Thus, due to employees focusing on a myriad of tasks, productivity issues are quite common.

5. Health issues

Computer network provides access to a gamut of services including entertainment, gaming, and movies. These result in making you addicted to the content and thus results in overuse of these services. This excessive screen time makes you feel lethargic, causes eye strain and body pain.

Applications of Computer Network:

- **Marketing and sales:** – Computer networks are widely used in both marketing sales firms. These are used by marketing professionals to collect, exchange, and analyzes data relating to customer requirements and product development cycles. Teleshopping is also important part of sales applications that use order-entry computers or telephones connected to an order-processing network, and on-line reservation services for hotels airline and so on.
- **Manufacturing:** – Now days, computer networks are used in a several aspects of manufacturing, including the manufacturing process itself. Two applications which use a network to provide necessary services are computer-assisted manufacturing (CAM) and computer –assisted designing (CAD) both of which permit multiple users to work on a project simultaneously.
- **Financial Services:** – In Present, Financial services are completely dependent on computer networks. Main applications are credit history searches, foreign exchange and investment services, and Electronic Funds Transfer (EFT) that permits a user to transfer money without going into bank.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- Teleconferencing: – With The help of teleconferencing conferences are possible to occur without the participants being in the same place. Applications include simple text conferencing, voice conferencing, and video conferencing.
- Cable Television:-Future Services provided by cable television network can include video on request, as well as the same information, financial and communications services currently provided by the telephone companies and computer networks.
- Information Services:- Network information services include bulletin boards and data banks. A World Wide Web site offering the technical specifications for a new product is an information service.
- Electronic Messaging:– Electronic mail (e-mail) is the most widely used network application.
- Electronic Data Interchange (EDI):– EDI permits business information to be transferred without using paper.
- Directory services: – By using directory services, it is possible to store the last of files in a central location to speed worldwide search operations.
- Cellular Telephone: – In the past, two parties desiring to use the services of the telephone company had to be linked by a fixed physical connection. But, in present cellular network make it possible to maintain wireless phone connections even while travelling over large distances.

2. Network Hardware:

It is described based on transmission technology and scalability.

Transmission Technology:

Broadly speaking, there are two types of transmission technology that are in widespread use: broadcast links and point-to-point links.

Point-to-point links connect individual pairs of machines. To go from the source to the destination on a network made up of point-to-point links, short messages, called packets in certain contexts, may have to first visit one or more intermediate machines.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Broadcast network is the communication channel that is shared by all the machines on the network; packets sent by any machine are received by all the others. An address field within each packet specifies the intended recipient. Upon receiving a packet, a machine checks the address field. If the packet is intended for the receiving machine, that machine processes the packet; if the packet is intended for some other machine, it is just ignored.

Broadcast systems usually also allow the possibility of addressing a packet to all destinations by using a special code in the address field. When a packet with this code is transmitted, it is received and processed by every machine on the network. This mode of operation is called broadcasting. Some broadcast systems also support transmission to a subset of the machines, which known as multicasting.

Point-to-point transmission with exactly one sender and exactly one receiver is sometimes called unicasting.

Scalability:

Network are categorised based on scale.

1 m	Square meter	Personal Area Network
10 m	Room	Local Area Network
100 m	Building	
1 km	Campus	
10 km	City	Metro politician Area Network
100 km	Country	Wide Area Network
1000 km	Continent	
10,000 km	Planet	Internet

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Personal Area Networks:

PANs (Personal Area Networks) let devices communicate over the range of a person. A common example is a wireless network that connects a computer with its peripherals. Almost every computer has an attached monitor, keyboard, mouse, and printer. Some companies got together to design a short-range wireless network called Bluetooth to connect these components without wires. The idea is that if your devices have Bluetooth, then you need no cables.

Local Area Networks:

A LAN is a privately owned network that operates within and nearby a single building like a home, office or factory. LANs are widely used to connect personal computers and consumer electronics to let them share resources (e.g., printers) and exchange information. When LANs are used by companies, they are called enterprise networks.

Wireless LANs are very popular these days, especially in homes, older office buildings, cafeterias, and other places where it is too much trouble to install cables. In these systems, every computer has a radio modem and an antenna that it uses to communicate with other computers.

This device, called an AP (Access Point), wireless router, or base station, relays packets between the wireless computers and also between them and the Internet. Being the AP is like being the popular kid at school because everyone wants to talk to you. However, if other computers are close enough, they can communicate directly with one another in a peer-to-peer configuration. There is a standard for wireless LANs called IEEE 802.11, popularly known as WiFi, which has become very widespread.

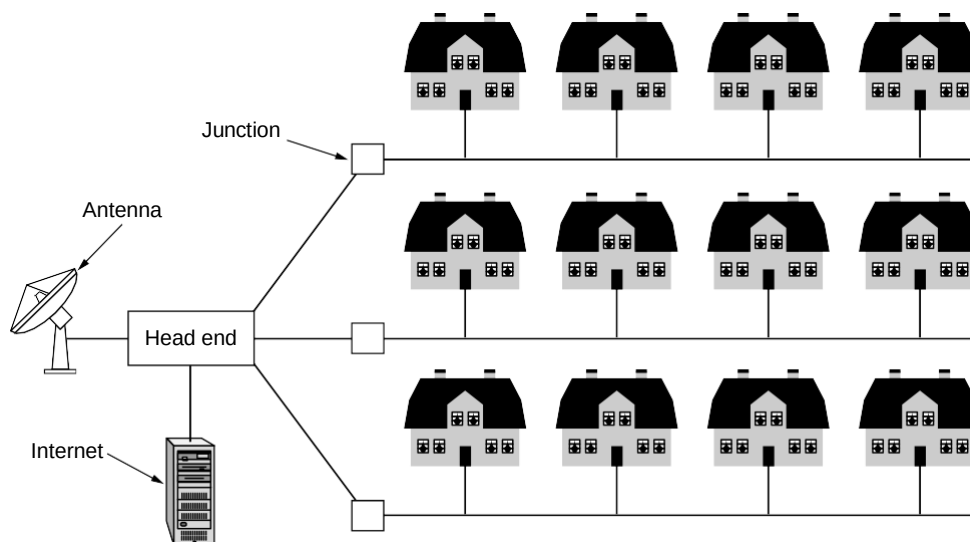
Metropolitan Area Networks

A MAN (Metropolitan Area Network) covers a city. The best-known examples of MANs are the cable television networks available in many cities. These systems grew from earlier community antenna systems used in areas with poor

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

over-the-air television reception. In those early systems, a large antenna was placed on top of a nearby hill and a signal was then piped to the subscribers' houses.

Internet access have resulted in another MAN, which has been standardized as IEEE 802.16 and is popularly known as WiMAX.



Wide Area Network:

It spans a large geographical area, often a country or continent. In most WANs, the subnet consists of two distinct components: transmission lines and switching elements. Transmission lines move bits between machines. They can be made of copper wire, optical fiber, or even radio links. Most companies do not have transmission lines lying about, so instead they lease the lines from a telecommunications company. Switching elements, or just switches, are specialized computers that connect two or more transmission lines. When data arrive on an incoming line, the switching element must choose an outgoing line on which to forward them. These switching computers have been called by various names in the past; the name router is now most commonly used.

Internetworks:

Many networks exist in the world, often with different hardware and software.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

People connected to one network often want to communicate with people attached to a different one. The fulfillment of this desire requires that different, and frequently incompatible, networks be connected. A collection of interconnected networks is called an internetwork or internet. These terms will be used in a generic sense, in contrast to the worldwide Internet (which is one specific internet), which we will always capitalize. The Internet uses ISP networks to connect enterprise networks, home networks, and many other networks.

3. NETWORK SOFTWARE

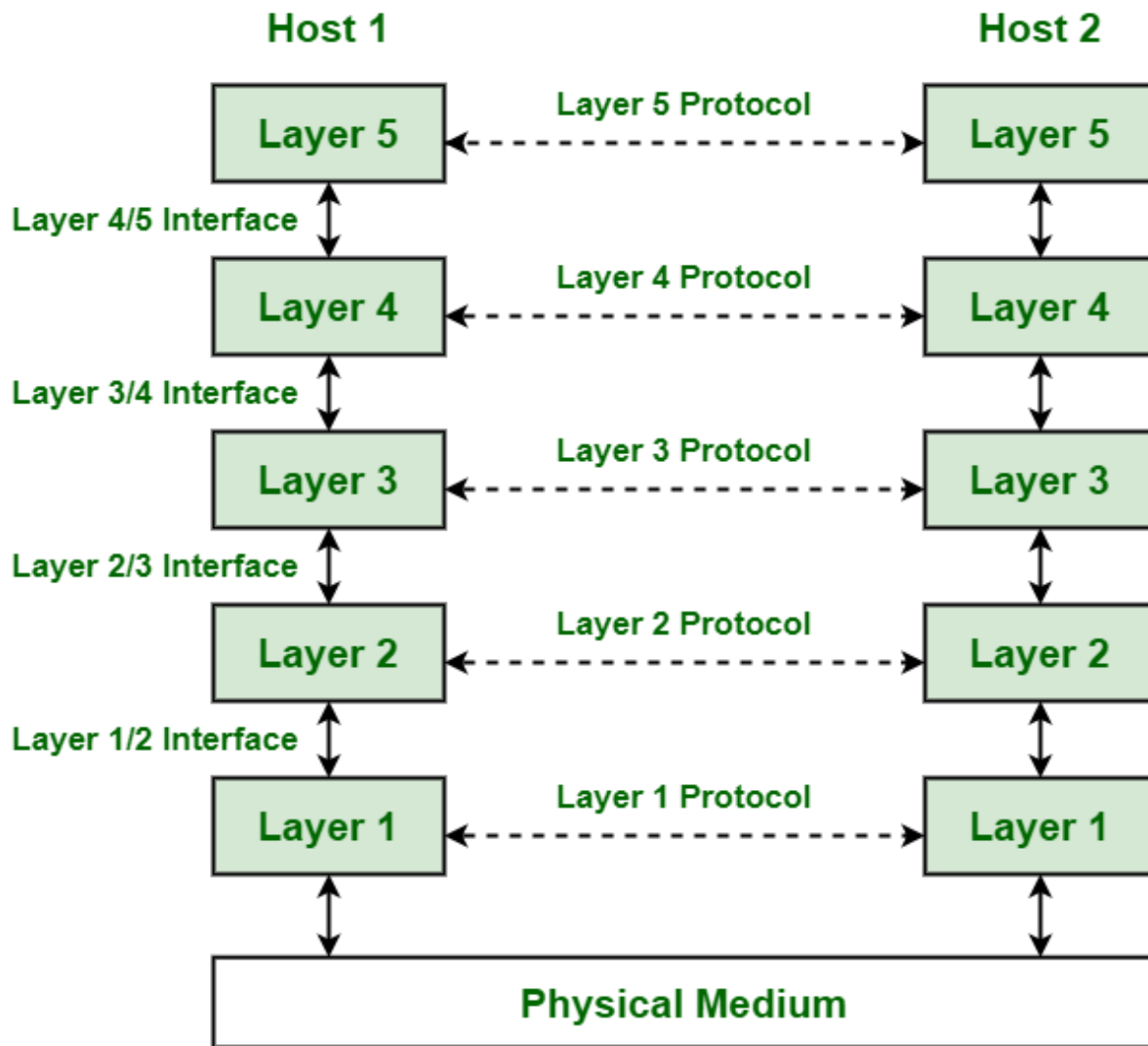
Protocol Hierarchies

To reduce their design complexity, most networks are organized as a stack of layers or levels, each one built upon the one below it. The number of layers, the name of each layer, the contents of each layer, and the function of each layer differ from network to network. The purpose of each layer is to offer certain services to the higher layers while shielding those layers from the details of how the offered services are actually implemented. Basically, a protocol is an agreement between the communicating parties on how communication is to proceed.

A five-layer network is illustrated below

The entities comprising the corresponding layers on different machines are called peers. The peers may be software processes, hardware devices, or even human beings. In other words, it is the peers that communicate by using the protocol to talk to each other.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE



Physical Hierarchies

The diagram shows communication between Host 1 and Host 2. The data stream is passed through a number of layers from one host to other. Virtual communication is represented using dotted lines between peer layers. Physical communication is represented using solid arrows between adjacent layers. Through physical medium, actual communication occurs. The layers at same level are commonly known as peers. The peer basically has a set of communication protocols. An interface is present between each of layers that are used to explain services provided by lower layer to higher layer.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Connection-Oriented Versus Connectionless Service:

Connection-oriented service is modelled after the telephone system. To talk to someone, you pick up the phone, dial the number, talk, and then hang up. Similarly, to use a connection-oriented network service, the service user first establishes a connection, uses the connection, and then releases the connection. The essential aspect of a connection is that it acts like a tube: the sender pushes objects (bits) in at one end, and the receiver takes them out at the other end.

In contrast to connection-oriented service, connectionless service is modeled after the postal system. Each message (letter) carries the full destination address, and each one is routed through the intermediate nodes inside the system independent of all the subsequent messages. There are different names for messages in different contexts; a packet is a message at the network layer. When the intermediate nodes receive a message in full before sending it on to the next node, this is called store-and-forward switching. The alternative, in which the onward transmission of a message at a node starts before it is completely received by the node, is called cut-through switching.

Service Primitives

A service is formally specified by a set of primitives (operations) available to user processes to access the service. These primitives tell the service to perform some action or report on an action taken by a peer entity. If the protocol stack is located in the operating system, as it often is, the primitives are normally system calls. These calls cause a trap to kernel mode, which then turns control of the machine over to the operating system to send the necessary packets. The set of primitives available depends on the nature of the service being provided. The primitives for connection-oriented service are different from those of connectionless service.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Primitive	Meaning
LISTEN	Block waiting for an incoming connection
CONNECT	Establish a connection with a waiting peer
RECEIVE	Block waiting for an incoming message
SEND	Send a message to the peer
DISCONNECT	Terminate a connection

First, the server executes LISTEN to indicate that it is prepared to accept incoming connections. A common way to implement LISTEN is to make it a blocking system call. After executing the primitive, the server process is blocked until a request for connection appears.

Next, the client process executes CONNECT to establish a connection with the server. The CONNECT call needs to specify who to connect to, so it might have a parameter giving the server's address.

The next step is for the server to execute RECEIVE to prepare to accept the first request. Normally, the server does this immediately upon being released from the LISTEN, before the acknowledgement can get back to the client. The RECEIVE call blocks the server.

Then the client executes SEND to transmit its request followed by the execution of RECEIVE to get the reply. The arrival of the request packet at the server machine unblocks the server so it can handle the request. After it has done the work, the server uses SEND to return the answer to the client. The arrival of this packet unblocks the client, which can now inspect the answer. If the client has additional requests, it can make them now.

When the client is done, it executes DISCONNECT to terminate the connection. Usually, an initial DISCONNECT is a blocking call, suspending the client and sending a packet to the server saying that the connection is no longer needed. When the server gets the

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE
packet, it also issues a DISCONNECT of its own, acknowledging the client and releasing the connection.

4. Reference Model

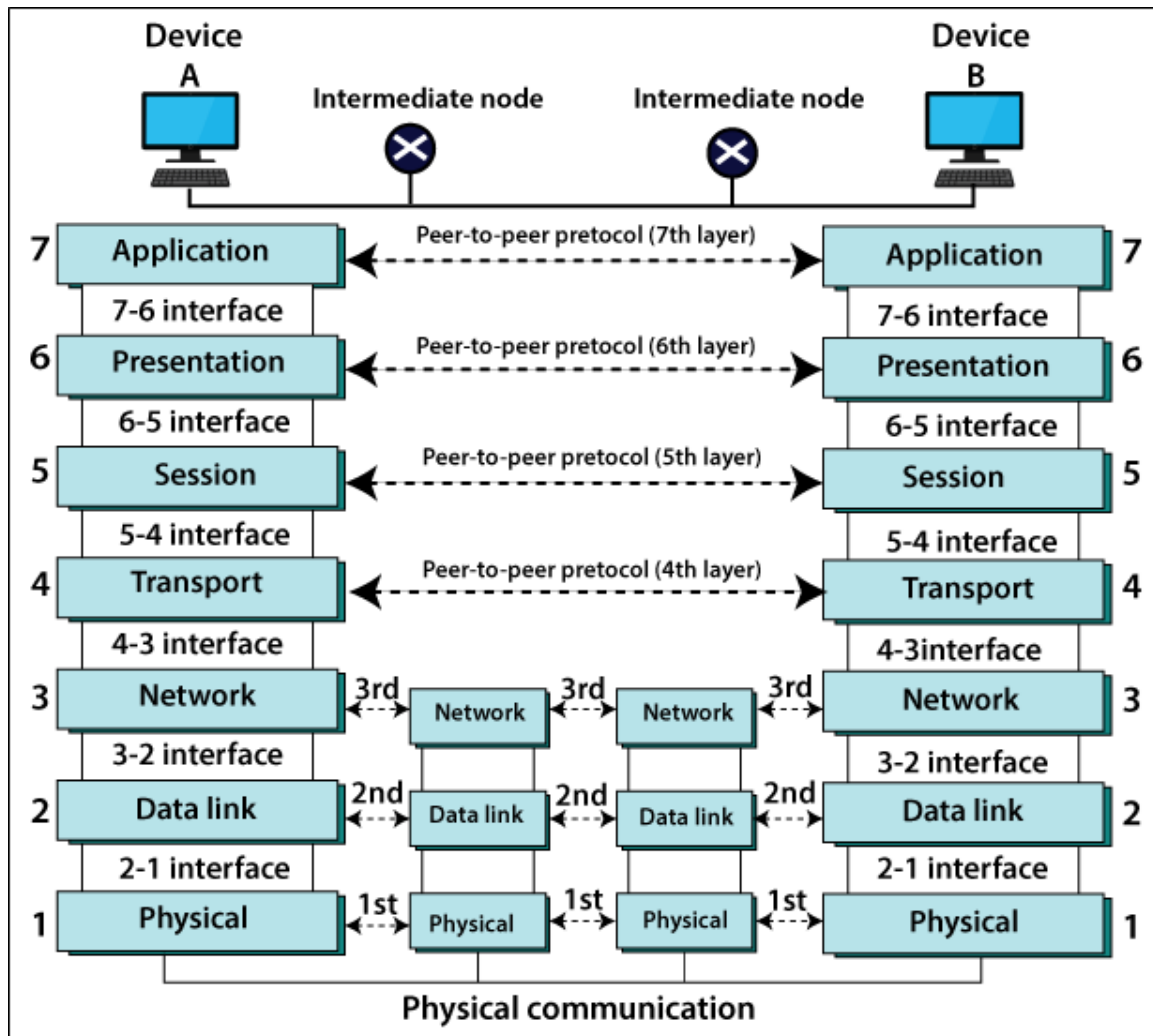
1. The OSI Reference Model

This model is based on a proposal developed by the International Standards Organization (ISO). The model is called the ISO OSI (Open Systems Interconnection) Reference Model because it deals with connecting open systems—that is, systems that are open for communication with other systems.

The OSI model has seven layers. The principles that were applied to arrive at the seven layers can be briefly summarized as follows:

1. A layer should be created where a different abstraction is needed.
2. Each layer should perform a well-defined function.
3. The function of each layer should be chosen with an eye toward defining internationally standardized protocols
4. The layer boundaries should be chosen to minimize the information flow across the interfaces.
5. The number of layers should be large enough that distinct functions need not be thrown together in the same layer out of necessity and small enough that the architecture does not become unwieldy.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE



Physical layer

The main functionality of the physical layer is to transmit the individual bits from one node to another node.

- It is the lowest layer of the OSI model.
- It establishes, maintains and deactivates the physical connection.
- It specifies the mechanical, electrical and procedural network interface specifications.

Functions of a Physical layer:

- Line Configuration: It defines the way how two or more devices can be connected physically.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- Data Transmission: It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.
- Topology: It defines the way how network devices are arranged.
- Signals: It determines the type of the signal used for transmitting the information.

2) Data-Link Layer

- This layer is responsible for the error-free transfer of data frames.
- It defines the format of the data on the network.
- It provides a reliable and efficient communication between two or more devices.
- It is mainly responsible for the unique identification of each device that resides on a local network.
- It contains two sub-layers:
 - Logical Link Control Layer
 - It is responsible for transferring the packets to the Network layer of the receiver that is receiving.
 - It identifies the address of the network layer protocol from the header.
 - It also provides flow control.
 - Media Access Control Layer
 - A Media access control layer is a link between the Logical Link Control layer and the network's physical layer.
 - It is used for transferring the packets over the network.

Functions of the Data-link layer

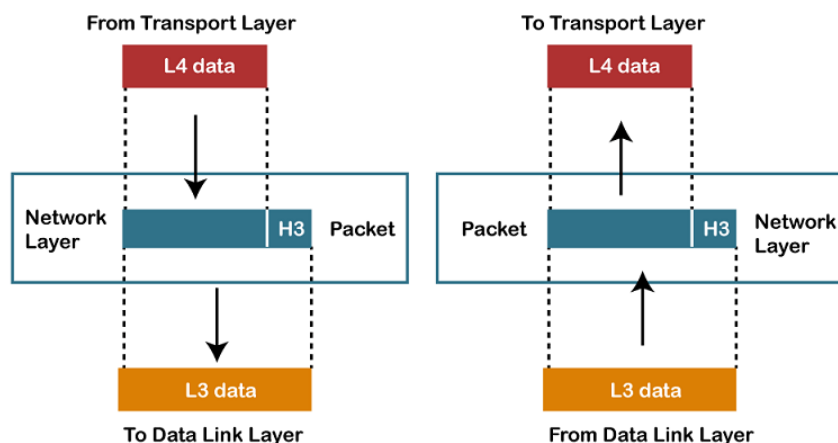
- Framing: The data link layer translates the physical's raw bit stream into packets known as Frames. The Data link layer adds the header and trailer to the frame.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

The header which is added to the frame contains the hardware destination and source address.

- Physical Addressing: The Data link layer adds a header to the frame that contains a destination address. The frame is transmitted to the destination address mentioned in the header.
- Flow Control: Flow control is the main functionality of the Data-link layer. It is the technique through which the constant data rate is maintained on both the sides so that no data get corrupted. It ensures that the transmitting station such as a server with higher processing speed does not exceed the receiving station, with lower processing speed.
- Error Control: Error control is achieved by adding a calculated value CRC (Cyclic Redundancy Check) that is placed to the Data link layer's trailer which is added to the message frame before it is sent to the physical layer. If any error seems to occur, then the receiver sends the acknowledgment for the retransmission of the corrupted frames.
- Access Control: When two or more devices are connected to the same communication channel, then the data link layer protocols are used to determine which device has control over the link at a given time.

Network Layer



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- It is a layer 3 that manages device addressing, tracks the location of devices on the network.
- It determines the best path to move data from source to the destination based on the network conditions, the priority of service, and other factors.
- The Data link layer is responsible for routing and forwarding the packets.
- Routers are the layer 3 devices, they are specified in this layer and used to provide the routing services within an internetwork.
- The protocols used to route the network traffic are known as Network layer protocols. Examples of protocols are IP and Ipv6.

Functions of Network Layer:

- Internetworking: An internetworking is the main responsibility of the network layer. It provides a logical connection between different devices.
- Addressing: A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet.
- Routing: Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.
- Packetizing: A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).

4) Transport Layer

- The Transport layer is a Layer 4 ensures that messages are transmitted in the order in which they are sent and there is no duplication of data.
- The main responsibility of the transport layer is to transfer the data completely.
- It receives the data from the upper layer and converts them into smaller units known as segments.
- This layer can be termed as an end-to-end layer as it provides a point-to-point connection between source and destination to deliver the data reliably.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

The two protocols used in this layer are:

- Transmission Control Protocol
 - It is a standard protocol that allows the systems to communicate over the internet.
 - It establishes and maintains a connection between hosts.
 - When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments. Each segment travels over the internet using multiple routes, and they arrive in different orders at the destination. The transmission control protocol reorders the packets in the correct order at the receiving end.
- User Datagram Protocol
 - User Datagram Protocol is a transport layer protocol.
 - It is an unreliable transport protocol as in this case receiver does not send any acknowledgment when the packet is received, the sender does not wait for any acknowledgment. Therefore, this makes a protocol unreliable.

5) Session Layer

- It is a layer 3 in the OSI model.
- The Session layer is used to establish, maintain and synchronizes the interaction between communicating devices.

Functions of Session layer:

- Dialog control: Session layer acts as a dialog controller that creates a dialog between two processes or we can say that it allows the communication between two processes which can be either half-duplex or full-duplex.
- Synchronization: Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and recovery.

6) Presentation Layer

- A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.
- It acts as a data translator for a network.
- This layer is a part of the operating system that converts the data from one presentation format to another format.
- The Presentation layer is also known as the syntax layer.

Functions of Presentation layer:

- Translation: The processes in two systems exchange the information in the form of character strings, numbers and so on. Different computers use different encoding methods, the presentation layer handles the interoperability between the different encoding methods. It converts the data from sender-dependent format into a common format and changes the common format into receiver-dependent format at the receiving end.
- Encryption: Encryption is needed to maintain privacy. Encryption is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.
- Compression: Data compression is a process of compressing the data, i.e., it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

7) Application Layer

- An application layer serves as a window for users and application processes to access network service.
- It handles issues such as network transparency, resource allocation, etc.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- An application layer is not an application, but it performs the application layer functions.
- This layer provides the network services to the end-users.

Functions of Application layer:

- File transfer, access, and management (FTAM): An application layer allows a user to access the files in a remote computer, to retrieve the files from a computer and to manage the files in a remote computer.
- Mail services: An application layer provides the facility for email forwarding and storage.
- Directory services: An application provides the distributed database sources and is used to provide that global information about various objects.

TCP/IP model:

TCP/IP stands for Transmission Control Protocol/Internet Protocol and is a suite of communication protocols used to interconnect network devices on the internet. TCP/IP is also used as a communications protocol in a private computer network (an intranet or extranet).

TCP/IP specifies how data is exchanged over the internet by providing end-to-end communications that identify how it should be broken into packets, addressed, transmitted, routed and received at the destination. TCP/IP requires little central management and is designed to make networks reliable with the ability to recover automatically from the failure of any device on the network. TCP defines how applications can create channels of communication across a network. It also manages how a message is assembled into smaller packets before they are then transmitted over the internet and reassembled in the right order at the destination address.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

IP defines how to address and route each packet to make sure it reaches the right destination. Each gateway computer on the network checks this IP address to determine where to forward the message.

A subnet mask tells a computer, or other network device, what portion of the IP address is used to represent the network and what part is used to represent hosts, or other computers, on the network.

TCP/IP uses the client-server model of communication in which a user or machine (a client) is provided a service, like sending a webpage, by another computer (a server) in the network.

TCP/IP model has four layers as network interface layer, internet layer, transport layer and application layer.

Network Access Layer

- A network layer is the lowest layer of the TCP/IP model.
- A network layer is the combination of the Physical layer and Data Link layer defined in the OSI reference model.
- It defines how the data should be sent physically through the network.
- This layer is mainly responsible for the transmission of the data between two devices on the same network.
- The functions carried out by this layer are encapsulating the IP datagram into frames transmitted by the network and mapping of IP addresses into physical addresses.
- The protocols used by this layer are ethernet, token ring, FDDI, X.25, frame relay.

Internet Layer

- An internet layer is the second layer of the TCP/IP model.
- An internet layer is also known as the network layer.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- The main responsibility of the internet layer is to send the packets from any network, and they arrive at the destination irrespective of the route they take.

Following are the protocols used in this layer are:

IP Protocol: IP protocol is used in this layer, and it is the most significant part of the entire TCP/IP suite.

Following are the responsibilities of this protocol:

- IP Addressing: This protocol implements logical host addresses known as IP addresses. The IP addresses are used by the internet and higher layers to identify the device and to provide internetwork routing.
- Host-to-host communication: It determines the path through which the data is to be transmitted.
- Data Encapsulation and Formatting: An IP protocol accepts the data from the transport layer protocol. An IP protocol ensures that the data is sent and received securely, it encapsulates the data into message known as IP datagram.
- Fragmentation and Reassembly: The limit imposed on the size of the IP datagram by data link layer protocol is known as Maximum Transmission unit (MTU). If the size of IP datagram is greater than the MTU unit, then the IP protocol splits the datagram into smaller units so that they can travel over the local network. Fragmentation can be done by the sender or intermediate router. At the receiver side, all the fragments are reassembled to form an original message.
- Routing: When IP datagram is sent over the same local network such as LAN, MAN, WAN, it is known as direct delivery. When source and destination are on the distant network, then the IP datagram is sent indirectly. This can be accomplished by routing the IP datagram through various devices such as routers.

ARP Protocol

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- ARP stands for Address Resolution Protocol.
- ARP is a network layer protocol which is used to find the physical address from the IP address.

ICMP Protocol

- ICMP stands for Internet Control Message Protocol.
- It is a mechanism used by the hosts or routers to send notifications regarding datagram problems back to the sender.
- A datagram travels from router-to-router until it reaches its destination. If a router is unable to route the data because of some unusual conditions such as disabled links, a device is on fire or network congestion, then the ICMP protocol is used to inform the sender that the datagram is undeliverable.
- An ICMP protocol mainly uses two terms:
 - ICMP Test: ICMP Test is used to test whether the destination is reachable or not.
 - ICMP Reply: ICMP Reply is used to check whether the destination device is responding or not.

Transport Layer

The transport layer is responsible for the reliability, flow control, and correction of data which is being sent over the network.

The two protocols used in the transport layer are User Datagram protocol and Transmission control protocol.

- User Datagram Protocol (UDP)
 - It provides connectionless service and end-to-end delivery of transmission.
 - It is an unreliable protocol as it discovers the errors but not specify the error.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- User Datagram Protocol discovers the error, and ICMP protocol reports the error to the sender that user datagram has been damaged.
- UDP consists of the following fields:
Source port address: The source port address is the address of the application program that has created the message.
Destination port address: The destination port address is the address of the application program that receives the message.
- Transmission Control Protocol (TCP)
 - It provides a full transport layer services to applications.
 - It creates a virtual circuit between the sender and receiver, and it is active for the duration of the transmission.
 - TCP is a reliable protocol as it detects the error and retransmits the damaged frames. Therefore, it ensures all the segments must be received and acknowledged before the transmission is considered to be completed and a virtual circuit is discarded.
 - At the sending end, TCP divides the whole message into smaller units known as segment, and each segment contains a sequence number which is required for reordering the frames to form an original message.
 - At the receiving end, TCP collects all the segments and reorders them based on sequence numbers.

Application Layer

- An application layer is the topmost layer in the TCP/IP model.
- It is responsible for handling high-level protocols, issues of representation.
- This layer allows the user to interact with the application.
- When one application layer protocol wants to communicate with another application layer, it forwards its data to the transport layer.
- There is an ambiguity occurs in the application layer. Every application cannot be placed inside the application layer except those who interact with the

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE
communication system. For example: text editor cannot be considered in application layer while web browser using HTTP protocol to interact with the network where HTTP protocol is an application layer protocol.

Following are the main protocols used in the application layer:

- HTTP: HTTP stands for Hypertext transfer protocol. This protocol allows us to access the data over the world wide web. It transfers the data in the form of plain text, audio, video. It is known as a Hypertext transfer protocol as it has the efficiency to use in a hypertext environment where there are rapid jumps from one document to another.
- SNMP: SNMP stands for Simple Network Management Protocol. It is a framework used for managing the devices on the internet by using the TCP/IP protocol suite.
- SMTP: SMTP stands for Simple mail transfer protocol. The TCP/IP protocol that supports the e-mail is known as a Simple mail transfer protocol. This protocol is used to send the data to another e-mail address.
- DNS: DNS stands for Domain Name System. An IP address is used to identify the connection of a host to the internet uniquely. But, people prefer to use the names instead of addresses. Therefore, the system that maps the name to the address is known as Domain Name System.
- TELNET: It is an abbreviation for Terminal Network. It establishes the connection between the local computer and remote computer in such a way that the local terminal appears to be a terminal at the remote system.
- FTP: FTP stands for File Transfer Protocol. FTP is a standard internet protocol used for transmitting the files from one computer to another computer.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Comparison between TCP/IP and OSI:

Following are the differences between OSI and TCP/IP Reference Model –

OSI	TCP/IP
OSI represents Open System Interconnection.	TCP/IP model represents the Transmission Control Protocol / Internet Protocol.
OSI is a generic, protocol independent standard. It is acting as an interaction gateway between the network and the final-user.	TCP/IP model depends on standard protocols about which the computer network has created. It is a connection protocol that assigns the network of hosts over the internet.
The OSI model was developed first, and then protocols were created to fit the network architecture's needs.	The protocols were created first and then built the TCP/IP model.
It provides quality services.	It does not provide quality services.
The OSI model represents defines administration, interfaces and conventions. It describes clearly which layer provides services.	It does not mention the services, interfaces, and protocols.
The protocols of the OSI model are better unseen and can be returned with another appropriate protocol quickly.	The TCP/IP model protocols are not hidden, and we cannot fit a new protocol stack in it.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

OSI	TCP/IP
It is difficult as distinguished to TCP/IP.	It is simpler than OSI.
It provides both connection and connectionless oriented transmission in the network layer; however, only connection-oriented transmission in the transport layer.	It provides connectionless transmission in the network layer and supports connecting and connectionless-oriented transmission in the transport layer.
It uses a vertical approach.	It uses a horizontal approach.
The smallest size of the OSI header is 5 bytes.	The smallest size of the TCP/IP header is 20 bytes.
Protocols are unknown in the OSI model and are returned while the technology modifies.	In TCP/IP, returning protocol is not difficult.

5. Example Networks:

ARPANET stands for Advanced Research Projects Agency NET. ARPANET was first network which consisted of distributed control. It was first to implement TCP/IP protocols. It was basically beginning of Internet with use of these technologies. It was designed with a basic idea in mind that was to communicate with scientific users among an institute or university.

History of ARPANET:

ARPANET was introduced in the year 1969 by Advanced Research Projects Agency (ARPA) of US Department of Defence. It was established using a bunch of PCs at various colleges and sharing of information and messages was done. It was for playing as long separation diversions and individuals were asked to share their perspectives. In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

the year 1980, ARPANET was handed over to different military network, Defense Data Network.

Characteristics of ARPANET:

1. It is basically a type of WAN.
2. It used concept of Packet Switching Network.
3. It used Interface Message Processors(IMPs) for sub-netting.
4. ARPANETs software was split into two parts- a host and a subnet.

Advantages of ARPANET:

- ARPANET was designed to service even in a Nuclear Attack.
- It was used for collaborations through E-mails.
- It created an advancement in transfer of important files and data of defense.

Limitations of ARPANET:

- Increased number of LAN connections resulted in difficulty handling.
- It was unable to cope-up with advancement in technology.

NSFNET:

National Science Foundation Network. NSFNET is a wide area network started by the NSF (National Science Foundation) that handled a bulk of early Internet traffic. It went online in 1986 and during the late 1980s and early 1990s was a crucial backbone to ARPANET and the Internet. During 1990 and 1991, NSFNET was restructured and created a not-for-profit entity and a for-profit subsidiary for commercial development of the network.

On April 30, 1995, NSF removed the NSFNET from the Internet but continued as a separate network for further research and development. After being shut down, this allowed more commercial companies to operate and expand on the Internet and helped started the dot-com boom.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Internet:

Internet is the foremost important tool and the prominent resource that is being used by almost every person across the globe. It connects millions of computers, webpages, websites, and servers. Using the internet we can send emails, photos, videos, and messages to our loved ones. Or in other words, the Internet is a widespread interconnected network of computers and electronic devices(that support Internet). It creates a communication medium to share and get information online. If your device is connected to the Internet then only you will be able to access all the applications, websites, social media apps, and many more services. The Internet nowadays is considered the fastest medium for sending and receiving information.

The actual working of the internet takes place with the help of clients and servers. Here the client is a laptop that is directly connected to the internet and servers are the computers connected indirectly to the Internet and they are having all the websites stored in those large computers. These servers are connected to the internet with the help of ISP (Internet Service Providers) and will be identified with the IP address.

X.25:

X.25 is generally a protocol that was developed by Telecommunication Standardization Sector (ITU-T) of International Telecommunication Union. It usually allows various logical channels to make use of same physical line. It basically defines a series of documents particularly issued by ITU. These documents are also known as X.25 Recommendations. X.25 also supports various conversations by multiplexing packets and also with the help of virtual communication channels. X.25 basically encompasses or suits to the lower three layers of the Open System Interconnection (OSI) reference model for networking. These three protocol layers are :

1. Physical Layer
2. Frame Layer
3. Packet Layer

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

1. **Physical Layer :** This layer is basically concerned with electrical or signaling. The physical layer interface of X.25 also known as X.21 bis was basically derived from RS-232 interface for serial transmission. This layer provides various communication lines that transmit or transfer some electrical signals. X.21 implementer is usually required for linking.
2. **Data Link Layer :** Data link layer is also known as Frame Layer. This layer is an implementation or development of ISO High-Level Data Link Layer (HDLC) standard which is known as LAPB (Link Access Procedure Balanced). It also provides a communication link and transmission that is error-free among any two physically connected nodes or X.25 nodes. LAPB also allows DTE (Data Terminal Equipment) or DCE (Data Circuit-Terminating Equipment) simply to start or end a communication session or start data transmission. This layer is one of the most important and essential parts of X.25 Protocol. This layer also provides a mechanism for checking in each hop during the transmission. This service also ensures a bit-oriented, error-free, and also sequenced and ordered delivery of data frames or packets.
3. **Packet Layer :** Packet layer is also known as Network Layer protocol of X.25. This layer generally governs the end-to-end communications among various DTE devices. It also defines how to address and deliver X.25 packets among end nodes and switches on a network with the help of PVCs (Permanent Virtual Circuits) or SVCs (Switched Virtual Circuits). This layer also defines and explains the format of data packets and also the procedures for control and transmission of data frames. This layer is also responsible for establishing a connection, transmitting data frames or packets, ending or terminating a connection, error and flow control, transmitting data packets over external virtual circuits.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Frame Relay:

Frame Relay is a packet-switching network protocol that is designed to work at the data link layer of the network. It is used to connect Local Area Networks (LANs) and transmit data across Wide Area Networks (WANs). It is a better alternative to a point-to-point network for connecting multiple nodes that require separate dedicated links to be established between each pair of nodes. It allows transmission of different size packets and dynamic bandwidth allocation. Also, it provides a congestion control mechanism to reduce the network overheads due to congestion. It does not have an error control and flow management mechanism.

Working:

Frame relay switches set up virtual circuits to connect multiple LANs to build a WAN. Frame relay transfers data between LANs across WAN by dividing the data in packets known as frames and transmitting these packets across the network. It supports communication with multiple LANs over the shared physical links or private lines.

Frame relay network is established between Local Area Networks (LANs) border devices such as routers and service provider network that connects all the LAN networks. Each LAN has an access link that connects routers of LAN to the service provider network terminated by the frame relay switch. The access link is the private physical link used for communication with other LAN networks over WAN. The frame relay switch is responsible for terminating the access link and providing frame relay services.

For data transmission, LAN's router (or other border device linked with access link) sends the data packets over the access link. The packet sent by LAN is examined by a frame relay switch to get the Data Link Connection Identifier (DLCI) which indicates the destination of the packet. Frame relay switch already has the information about addresses of the LANs connected to the network hence it identifies the destination LAN by looking at DLCI of the data packet. DLCI basically identifies the virtual circuit (i.e. logical path between nodes that doesn't really exist) between source and destination network. It configures and transmits the packet to frame relay switch of

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE
destination LAN which in turn transfers the data packet to destination LAN by sending it over its respective access link. Hence, in this way, a LAN is connected with multiple other LANs by sharing a single physical link for data transmission.

6.Transmission Media:

Guided transmission media:

Guided Media

It is defined as the physical medium through which the signals are transmitted. It is also known as Bounded media.

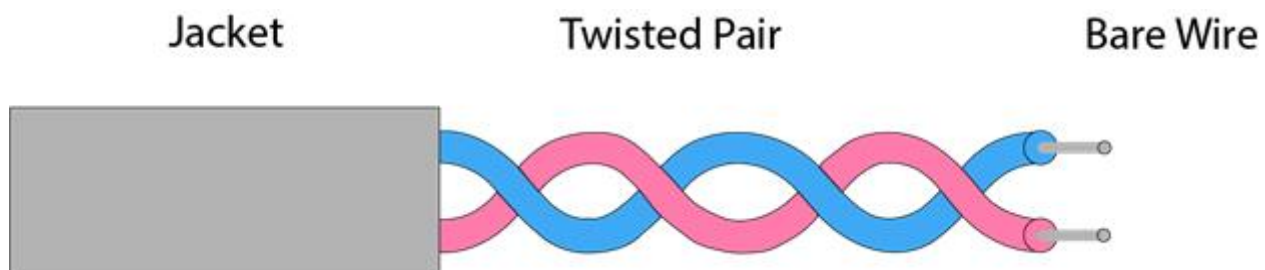
Types Of Guided media:

Twisted pair:

Twisted pair is a physical media made up of a pair of cables twisted with each other. A twisted pair cable is cheap as compared to other transmission media. Installation of the twisted pair cable is easy, and it is a lightweight cable. The frequency range for twisted pair cable is from 0 to 3.5KHz.

A twisted pair consists of two insulated copper wires arranged in a regular spiral pattern.

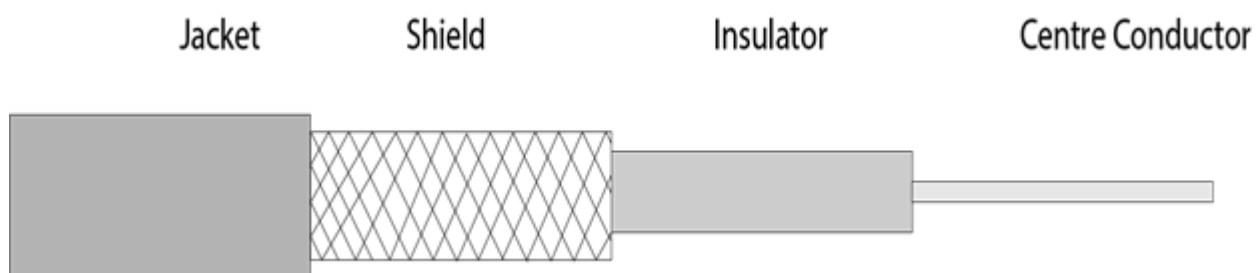
The degree of reduction in noise interference is determined by the number of turns per foot. Increasing the number of turns per foot decreases noise interference.



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Coaxial Cable

- Coaxial cable is very commonly used transmission media, for example, TV wire is usually a coaxial cable.
- The name of the cable is coaxial as it contains two conductors parallel to each other.
- It has a higher frequency as compared to Twisted pair cable.
- The inner conductor of the coaxial cable is made up of copper, and the outer conductor is made up of copper mesh. The middle core is made up of non-conductive cover that separates the inner conductor from the outer conductor.
- The middle core is responsible for the data transferring whereas the copper mesh prevents from the EMI(Electromagnetic interference).



Advantages Of Coaxial cable:

- The data can be transmitted at high speed.
- It has better shielding as compared to twisted pair cable.
- It provides higher bandwidth.

Disadvantages Of Coaxial cable:

- It is more expensive as compared to twisted pair cable.
- If any fault occurs in the cable causes the failure in the entire network.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Fibre Optic

- Fibre optic cable is a cable that uses electrical signals for communication.
- Fibre optic is a cable that holds the optical fibres coated in plastic that are used to send the data by pulses of light.
- The plastic coating protects the optical fibres from heat, cold, electromagnetic interference from other types of wiring.
- Fibre optics provide faster data transmission than copper wires.

Diagrammatic representation of fibre optic cable:



Basic elements of Fibre optic cable:

- Core: The optical fibre consists of a narrow strand of glass or plastic known as a core. A core is a light transmission area of the fibre. The more the area of the core, the more light will be transmitted into the fibre.
- Cladding: The concentric layer of glass is known as cladding. The main functionality of the cladding is to provide the lower refractive index at the core interface as to cause the reflection within the core so that the light waves are transmitted through the fibre.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- Jacket: The protective coating consisting of plastic is known as a jacket. The main purpose of a jacket is to preserve the fibre strength, absorb shock and extra fibre protection.

Advantages of fibre optic cable over copper:

- Greater Bandwidth: The fibre optic cable provides more bandwidth as compared copper. Therefore, the fibre optic carries more data as compared to copper cable.
- Faster speed: Fibre optic cable carries the data in the form of light. This allows the fibre optic cable to carry the signals at a higher speed.
- Longer distances: The fibre optic cable carries the data at a longer distance as compared to copper cable.

7. PSTN:

PSTN stands for Public Switched Telephone Network. It's a collection of the world's phone systems that use traditional circuit-switched telephone networks for public communications.

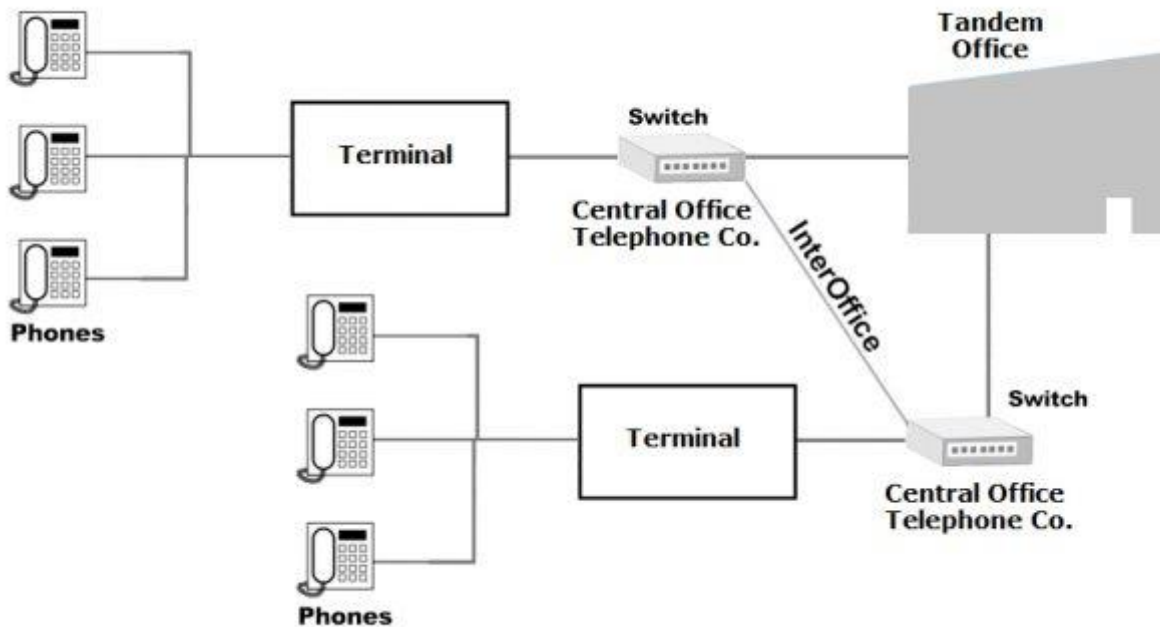
The telephone network is known by several names, such as PSTN, landlines, Plain Old Telephone Service (POTS), or fixed-line telephones. This system has been in general use since the late 1800s.

How the PSTN Works

Think of a PSTN as a combination of worldwide telephone networks, including telephone lines, fiber optic cables, switching centers, cellular networks, satellites, and cable systems. PSTN helps telephones communicate with each other.

Put simply, when you dial a phone number, your call moves through the network to reach its destination – and two phones connect. To fully understand how Plain Old

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE
Telephone Service (POTS) works, consider what happens when you dial a number from your phone.



From telephone handset to Central Office

Your telephone set converts your call (sound waves) into electrical signals and transmits them over network cables to a nearby exchange or terminal. The terminal collects these signals and determines where your voice (converted into electrical signals) goes next.

Central Office call routing

The call next goes to a tandem office (a regional hub that routes calls to distant central offices) or to a central office (for local calls). The central office routes the call through a fiber optic cable that converts the electrical signals into light pulses.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

From the Central Office to the terminating number:

The relevant center office picks up the light pulses, converts them into electrical signals, and sends them to the target terminal. The destination terminal forwards the call through sound waves to the appropriate telephone number.

It's natural to be confused about how PSTN works behind the scenes. Remember that it takes a few seconds for your call to reach its destination via fiber optic cables and a global network of switches. It's like changing buses to reach your final destination, only faster.

8. Mobile and telephone system:

Mobile Computing refers a technology that allows transmission of data, voice and video via a computer or any other wireless enabled device. It is free from having a connection with a fixed physical link. It facilitates the users to move from one physical location to another during communication.

Our Mobile communication tutorial includes all topics of mobile computing like its brief overview and history, evolution, classification, advantages and disadvantages, security issues, future trends etc.

Introduction of Mobile Computing

Mobile Computing is a technology that provides an environment that enables users to transmit data from one device to another device without the use of any physical link or cables.

In other words, you can say that mobile computing allows transmission of data, voice and video via a computer or any other wireless-enabled device without being

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

connected to a fixed physical link. In this technology, data transmission is done wirelessly with the help of wireless devices such as mobiles, laptops etc.

This is only because of Mobile Computing technology that you can access and transmit data from any remote locations without being present there physically. Mobile computing technology provides a vast coverage diameter for communication. It is one of the fastest and most reliable sectors of the computing technology field.

The concept of Mobile Computing can be divided into three parts:

- Mobile Communication
- Mobile Hardware
- Mobile Software

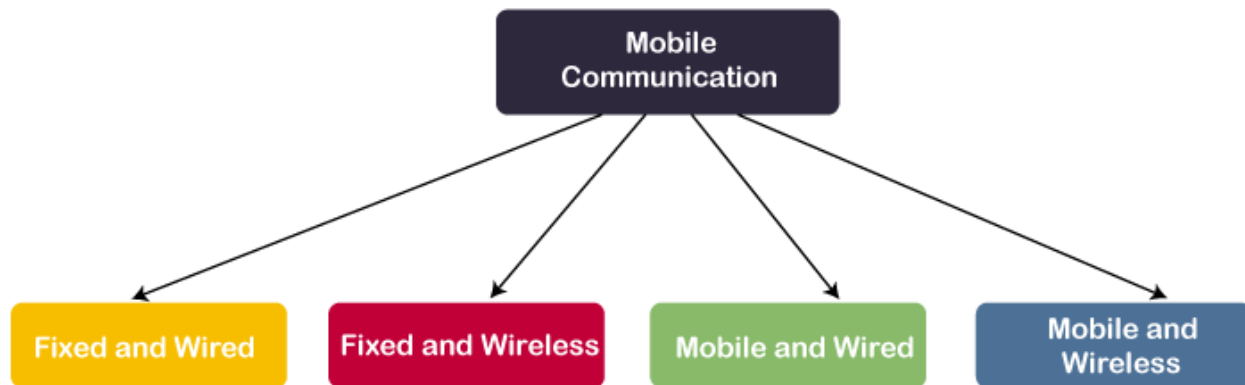
Mobile Communication

Mobile Communication specifies a framework that is responsible for the working of mobile computing technology. In this case, mobile communication refers to an infrastructure that ensures seamless and reliable communication among wireless devices. This framework ensures the consistency and reliability of communication between wireless devices. The mobile communication framework consists of communication devices such as protocols, services, bandwidth, and portals necessary to facilitate and support the stated services. These devices are responsible for delivering a smooth communication process.

Mobile communication can be divided in the following four types:

1. Fixed and Wired
2. Fixed and Wireless
3. Mobile and Wired
4. Mobile and Wireless

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE



Fixed and Wired: In Fixed and Wired configuration, the devices are fixed at a position, and they are connected through a physical link to communicate with other devices.

For Example, Desktop Computer.

Fixed and Wireless: In Fixed and Wireless configuration, the devices are fixed at a position, and they are connected through a wireless link to make communication with other devices.

For Example, Communication Towers, WiFi router

Mobile and Wired: In Mobile and Wired configuration, some devices are wired, and some are mobile. They altogether make communication with other devices.

For Example, Laptops.

Mobile and Wireless: In Mobile and Wireless configuration, the devices can communicate with each other irrespective of their position. They can also connect to any network without the use of any wired device.

For Example, WiFi Dongle.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Mobile Hardware:

Mobile hardware consists of mobile devices or device components that can be used to receive or access the service of mobility. Examples of mobile hardware can be smartphones, laptops, portable PCs, tablet PCs, Personal Digital Assistants, etc.



These devices are inbuilt with a receptor medium that can send and receive signals. These devices are capable of operating in full-duplex. It means they can send and receive signals at the same time. They don't have to wait until one device has finished communicating for the other device to initiate communications.

Mobile Software:

Mobile software is a program that runs on mobile hardware. This is designed to deal capably with the characteristics and requirements of mobile applications. This is the operating system for the appliance of mobile devices. In other words, you can say it the heart of the mobile systems. This is an essential component that operates the mobile device.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

Generations of Mobile technology:

0th Generation:

- Pre-cell phone mobile telephony technology, such as radio telephones some had in cars before the arrival of cell phones.
- Communication was possible through voice only.
- These mobile telephones were usually mounted in cars or trucks.

1G (1st Generation):

- First-time calling was introduced in mobile systems.
- It used analog signals.
- It used an FDD scheme and typically allocated a bandwidth of 25 Mhz.
- The coverage area was small.
- No roaming support between various operators.
- Low sound quality.
- Speed:- 2.4 Kbps.

2G (2nd Generation) :

- Shifted from analog to digital.
- It supported voice and SMS both.
- Supported all 4 sectors of the wireless industry namely Digital cellular, Mobile Data, PCS, WLAN,
- Moderate mobile data service.
- 2G WLAN provided a high data rate & large area coverage.
- Speed:- 64 Kbps.

2.5G came after 2G which used the concept of GPRS. Streaming was also introduced and mail services too. Then came 2.75G or EDGE which was faster in providing services than 2.5G. It gave faster internet speed up to 128kbps and also used edge connection.

3G (3rd Generation) :

- The Internet system was improved.
- Better system and capacity.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING – DATA SCIENCE

- Offers high-speed wireless internet.
- The connection used was UMTS and WCMA.
- Speed:- 2Mbps.

4G (4th Generation) :

- IP-based protocols.
- LTE (Long term evaluation) was mainly for the internet.
- Vo-LTE (Voice over LTE) is for both voice and the internet.
- Freedom and flexibility to select any desired service with reasonable QoS.
- High usability.
- Supports multimedia service at a low transmission cost.
- HD Quality Streaming.
- Speed:-100Mbps.

5G (5th Generation): It is yet to come in many countries but here are some notable points about 5G.

- Higher data rates.
- Connectivity will be more fast and more secure,
- Data Latency will be reduced to a great level.
- Massive network capacity.
- It is 30 times faster than 4G.
- There would be more flexibility in the network.
- Speed – 1Gbps